

Analytics and Types of Analytics

Analytics is the process of collecting, organizing, and analyzing data to make better business decisions.

Analytics helps convert this raw data into useful information for decision-making.

There are mainly **four types of analytics** used in business:

1. Descriptive Analytics
2. Diagnostic Analytics
3. Predictive Analytics
4. Prescriptive Analytics

1. Descriptive Analytics

Descriptive analytics explains **what happened in the past** using historical data. It summarizes data through reports, charts, and dashboards.

Example:

- A company checks last year's sales performance.

Tools Used

- Microsoft Excel
- Power BI
- Tableau

2. Diagnostic Analytics

Diagnostic analytics explains **why something happened**. It analyzes data deeply to find reasons for problems or success.

Examples

- Finding reasons for low sales
- Analyzing customer complaints

Advantages

1. Identifies business problems
2. Helps improve performance
3. Better decision-making

3. Predictive Analytics

Predictive analytics predicts **what may happen in the future** using historical data and statistical methods.

Features

- Uses machine learning and statistics
- Predicts future trends
- Helps in planning

Examples

- An online shopping company predicts festival season sales.
- Predicting stock market trends

4. Prescriptive Analytics

Prescriptive analytics suggests **what action should be taken** for the best result. It provides recommendations based on data analysis.

Examples

- Route suggestions in delivery services
 - Product recommendations in shopping apps
 - Netflix recommends movies based on user interest.
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Comparison of Types of Analytics

Type of Analytics	Main Question	Purpose
Descriptive Analytics	What happened?	Understand past data
Diagnostic Analytics	Why did it happen?	Find reasons
Predictive Analytics	What may happen?	Predict future
Prescriptive Analytics	What should be done?	Suggest actions

Applications of Analytics

1. Healthcare
2. Banking
3. Education
4. E-commerce
5. Manufacturing
6. Transportation

Analytics Lifecycle

The **Analytics Lifecycle** is a step-by-step process used in business analytics to collect, process, analyze, and interpret data for making better business decisions.

In modern businesses, huge amounts of data are generated every day. To convert this raw data into useful information, companies follow a systematic process called the analytics lifecycle.

The analytics lifecycle helps:

- Understand business problems
 - Analyze data properly
 - Make accurate decisions
 - Improve business performance
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Phases of Analytics Lifecycle

The analytics lifecycle mainly contains **six phases**:

1. Discovery
 2. Data Preparation
 3. Model Planning
 4. Model Building
 5. Communicate Results
 6. Operationalize
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1. Discovery Phase

In this phase, the business problem and project goals are identified like,

- What problem needs to be solved
- What data is required
- What resources are needed

Example

A shopping company wants to analyze why online sales are decreasing.

2. Data Preparation Phase

In this phase, data is collected, cleaned, and organized for analysis.

Raw data may contain:

- Missing values
- Errors
- Duplicate records

Activities

- Collecting data
- Cleaning data
- Integrating data
- Transforming data

Example: Removing duplicate customer records from a database.

3. Model Planning Phase

In this phase, suitable analytical methods and techniques are selected.

The team decides:

- Which statistical methods to use (Selecting algorithms, Choosing analysis techniques)

Example

Using regression analysis to predict future sales.

4. Model Building Phase

In this phase, the analytical model is created. The model is trained and tested using data to identify patterns and make predictions.

Activities

- Building models
- Testing models
- Validating results

Example

Creating a machine learning model for customer recommendation.

5. Communicate Results Phase

In this phase, the results and findings are presented to management or stakeholders.

Results are shown using:

- Charts
- Graphs
- Reports
- Dashboards

Example: Presenting sales analysis reports to company managers.

6. Operationalize Phase

In this phase, the analytical solution is implemented in real business operations. The model is monitored regularly to ensure proper performance.

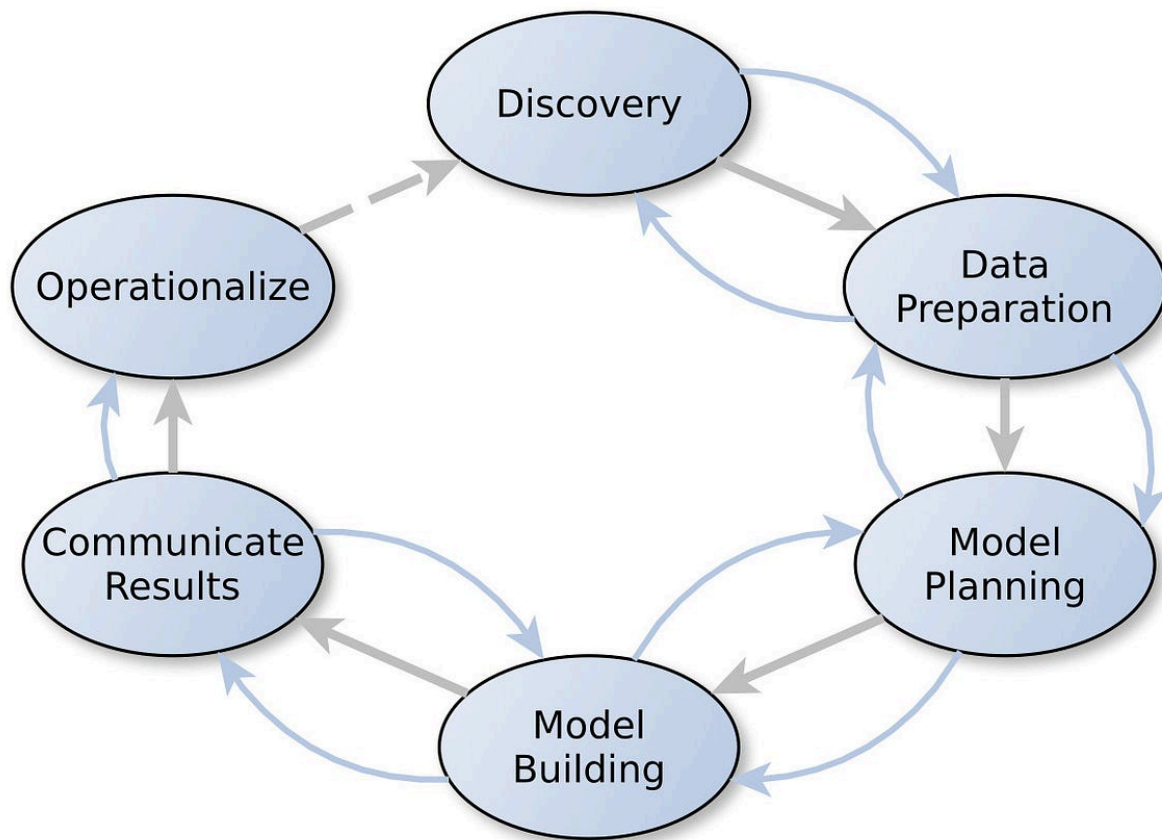
Activities

- Deploying models
- Monitoring performance
- Updating systems

Example

An e-commerce company implementing a recommendation system on its website.

Diagram of Analytics Lifecycle



Benefits of Analytics Lifecycle

1. Improves decision-making
2. Provides systematic data analysis
3. Reduces business risks
4. Increases operational efficiency
5. Helps achieve business goals
6. Improves prediction accuracy

Applications of Analytics Lifecycle

- Banking sector
- Healthcare industry
- E-commerce companies
- Manufacturing industries
- Education sector

Example:

- Amazon uses an analytics lifecycle for customer recommendations and delivery optimization.

Data Collection and Data Processing

- In modern businesses, a large amount of data is generated every day.
 - This data may come from customers, websites, machines, surveys, and transactions.
 - Before making decisions, businesses must:
 - Collect accurate data
 - Process the data properly
 - Proper data collection and processing help organizations:
 - Improve decision-making
 - Increase efficiency
 - Reduce errors
 - Understand customer needs
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Data Collection

Data collection is the first step in analytics and research. It is the process of gathering information from different sources for analysis and decision-making.

Types of Data Collection

Primary Data Collection	Secondary Data Collection
Primary data is collected directly from original sources.	Secondary data is collected from already existing sources.
Methods • Surveys • Interviews • Questionnaires • Observations	Sources • Books • Websites • Government reports • Research articles
Example A company conducting customer feedback surveys.	Example Using census data for market analysis.

Methods of Data Collection

1. Surveys

Information is collected from people using forms or questionnaires.

2. Interviews

Direct conversation is conducted with individuals to collect information.

3. Observation Method

Data is collected by observing behavior or activities.

4. Online Data Collection

Data is collected through websites, mobile apps, and social media platforms.

Example:

- Google collects user search data for analysis.

Importance of Data Collection

1. Helps in decision-making
 2. Improves business planning
 3. Identifies customer needs
 4. Supports research and analysis
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Data Processing

Data Processing is the process of converting raw data into meaningful and useful information.

Stages of Data Processing

1. Data Collection - Gathering raw data from various sources.

2. Data Preparation - Data is cleaned and organized.

Activities

- Removing duplicate data
- Correcting errors
- Handling missing values

3. Data Input - Prepared data is entered into a computer system.

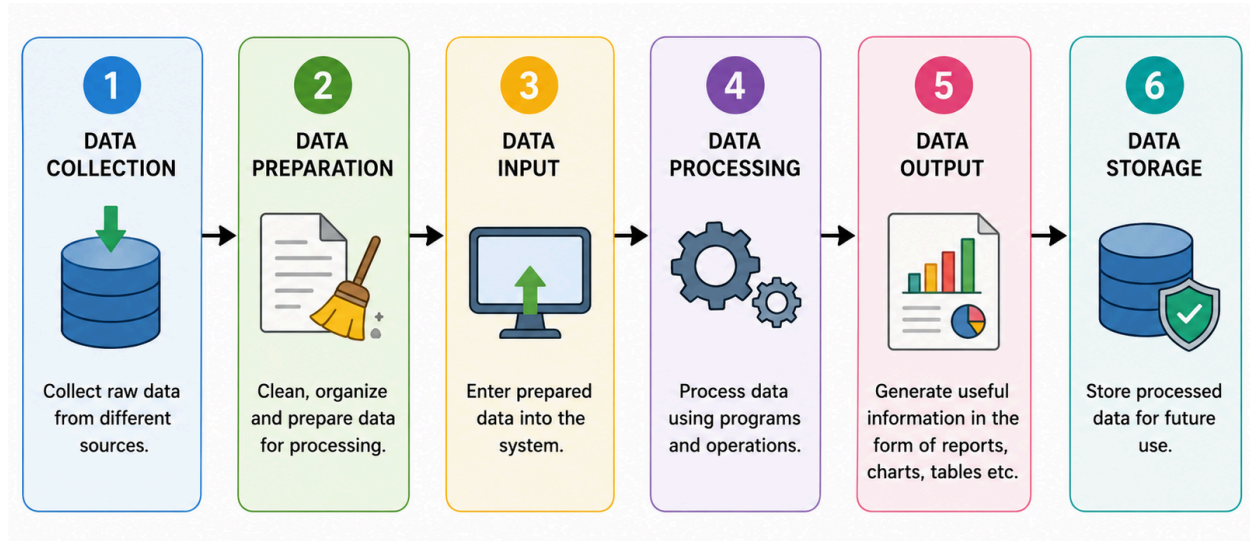
4. Data Processing - The computer processes data using algorithms and software.

5. Data Output - Processed information is presented in the form of:

- Reports
- Charts
- Tables
- Graphs

6. Data Storage - Processed data is stored for future use.

Diagram of Data Processing Steps



Advantages of Data Collection and Processing

1. Better business understanding
2. Faster decision-making
3. Accurate analysis
4. Better customer service

Applications

- Banking sector
- Education systems
- E-commerce companies
- Government organizations

Example:

- Amazon collects customer purchase data and processes it for recommendations.